3) a) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{2n+1}{3^n}$

$a_n = \frac{2n+1}{3^n}$, $\lim a_n = \frac{2n+1}{3^n} = 0$

$a_n \searrow$

$b_n = (-1)^{n-1} \in [0,1]$

$\sigma_1 = b_1 = 1$

$\sigma_2 = b_1 + b_2 = 1 - 1 = 0$

. - - - ..

A-D $\Rightarrow \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2n+1}{3^n}$ is conv.

b) $\sum_{n=1}^{\infty} \cos n \cdot \sin \frac{1}{n}$

Take $a_n = \sin \frac{1}{n} \searrow 0$

$b_n = \cos n \Rightarrow -1 \le b_n \le 1$

A-B $\Rightarrow \sum_{n=1}^{\infty} \cos n \cdot \sin \frac{1}{n}$ is conv.

c) $\sum_{n=1}^{\infty} \frac{\sin \cdot n \cdot \sin n^2}{\sqrt{n}}$

Take $a_n = \frac{1}{\sqrt{n}} \searrow 0$

$0 \le b_n = \sin \cdot n \cdot \sin n^2 \le 1$

A-B $\Rightarrow \sum_{n=1}^{\infty} \frac{\sin n \cdot \sin n^2}{\sqrt{n}}$ is conv.

d) $\sum_{n=1}^{\infty} \sin(\pi \sqrt{n^2+1}) = \sin(\pi \sqrt{n^2+1} - \pi n + \pi n)$

$$\sin\left(\pi\left(\sqrt{n^2+1}-n\right)+\pi n\right) = (-1)^n \sin\left(\frac{\pi}{\sqrt{n^2+1}+n}\right)$$

$$\sum_{n=1}^{\infty} (-1)^n \cdot \sin\left(\frac{\pi}{\sqrt{n^2+1}+n}\right)$$

Take $a_n = \sin\left(\frac{\pi}{\sqrt{n^2+1}+n}\right) \searrow 0$

$b_n = (-1)^n \in [-1, 0]$

| A-D $\Rightarrow \sum_{n=1}^{\infty} \sin(\pi\sqrt{n^2+1})$ is conv

4) $a_{n+1} = \ln(1+a_n)$

a) $\lim a_n = 0$

$a_{n+1} - a_n = \ln(1+a_n) - a_n = \ln\left(\frac{1+a_n}{e^{a_n}}\right)$

$a_1 = 1 > 0$

$a_2 = \ln 2 > 0$

- - - - -

$a_{n+1} > 0 \Rightarrow \ln(1+a_n) > 0 \mid \Rightarrow a_{n+1} > 0$

$a_n > 0$

$\Rightarrow a_n > 0$

$a_n > 0$

$e^{a_n} \geq 1$

$e^{a_n} > a_{n+1}$, pentru $a_n > 0$

$\Rightarrow a_{n+1} - a_n < 0$

$\Rightarrow a_n \in (0, 1]$ |

$a_n \searrow$

$\Rightarrow \lim a_n = 0$

b) $\sum_{n=1}^{\infty} a_n$ is div

D'alambert $\Rightarrow \lim \frac{a_{n+1}}{a_n} = \lim \frac{\ln(a_{n+1})}{a_n} = 1$ fails

5) $a_{n+1} = a_n^2 - a_n + 1$

$a_{n+1}^2 - 2a_{n+1} \cdot a_n + 1 = a_n^2 + 1$

$a_{n+1}^2 = -2a_{n+1} \cdot a_n$

$a_1 > 1$

$a_{n+1} - a_n = a_n^2 + 1 > 0 \Rightarrow$ P.C.

$\ell = \ell^2 - \ell + 1$

$\ell^2 - 2\ell + 1 = 0 \Rightarrow \ell = 1$ $\Big| \Rightarrow a_n \underset{n \to \infty}{\to} \infty$

$a_1 > 1$

$\sum_{n=1}^{\infty} \frac{1}{a_n} =$